An MPI-Parallel Unstructured Finite-Volume Solver for the Compressible Navier–Stokes Equations: Benchmark Report

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Abstract

We present a 2-D unstructured finite-volume solver for the compressible Navier–Stokes equations of a calorically perfect gas, with MPI domain decomposition based on METIS graph partitioning and neighbor-scoped halo exchange. The solver uses weighted least-squares linear reconstruction with Barth–Jespersen limiting, Roe/Rusanov Riemann fluxes, an LU-SGS implicit pseudo-time solver with analytic split-flux Jacobians, and a BDF2 dual-time integrator for transient cases. We report results for six NACA0012 cases (inviscid and laminar, Mach 0.15/0.8/2.0) and two cylinder cases (laminar, Re 20 steady and Re 200 vortex shedding).

1 Governing equations and gas model

We solve the conservative compressible Navier–Stokes equations in 2-D,

$$\frac{\partial \mathbf{U}}{\partial t} + \frac{\partial \mathbf{F}^i}{\partial x} + \frac{\partial \mathbf{G}^i}{\partial y} = \frac{\partial \mathbf{F}^v}{\partial x} + \frac{\partial \mathbf{G}^v}{\partial y}, \qquad \mathbf{U} = [\rho, \rho u, \rho v, \rho E]^T, \tag{1}$$

with inviscid fluxes

$$\mathbf{F}^i = [\rho u,\ \rho u^2 + p,\ \rho uv,\ u(\rho E + p)]^T, \quad \mathbf{G}^i = [\rho v,\ \rho uv,\ \rho v^2 + p,\ v(\rho E + p)]^T, \tag{2}$$

Newtonian viscous stresses and Fourier heat conduction,

$$\tau_{xx} = 2\mu u_x - \tfrac{2}{3}\mu(u_x + v_y), \quad \tau_{yy} = 2\mu v_y - \tfrac{2}{3}\mu(u_x + v_y), \quad \tau_{xy} = \mu(u_y + v_x), \quad \mathbf{q} = -k\nabla T, \quad k = \frac{\mu c_p}{Pr}, \tag{3}$$

and the calorically perfect gas model

$$p = (\gamma - 1)\rho e, \qquad E = e + \tfrac{1}{2}(u^2 + v^2), \qquad a = \sqrt{\gamma p/\rho}. \tag{4}$$

All cases use nondimensional freestream $\rho_\infty = 1$, $|\mathbf{V}_\infty| = 1$, $p_\infty = 1/(\gamma M_\infty^2)$, $\gamma = 1.4$, $R = 1$, $Pr = 0.72$; the laminar cases use constant viscosity $\mu = \rho_\infty U_\infty L_{ref}/Re$.

1.1 Nondimensionalization

All quantities are nondimensionalized with freestream density $\rho_\infty = 1$, speed $|V_\infty| = 1$, and reference length $L_{ref} = 1$; the freestream pressure is set by the Mach number, $p_\infty = \rho_\infty |V_\infty|^2/(\gamma M_\infty^2)$. Force coefficients use the dynamic pressure $q_\infty = \frac{1}{2}\rho_\infty |V_\infty|^2$ and the case reference area, $C_D = F_x/(q_\infty A_{ref})$, $C_L = F_y/(q_\infty A_{ref})$. The pressure and skin-friction coefficients are $C_p = (p - p_\infty)/q_\infty$ and $C_f = \tau_w/q_\infty$, with τ_w the tangential wall shear. Laminar viscosity is constant and matched to the case Reynolds number, $\mu = \rho_\infty |V_\infty| L_{ref}/Re$, with thermal conductivity $k = \mu c_p/Pr$.

2 Discretization

2.1 Finite-volume scheme

Cell-centered finite volumes on mixed triangular/quadrilateral meshes. For each cell i,

$$\frac{d\mathbf{U}_i}{dt} = -\frac{1}{V_i} \sum_{f \in \partial i} \left(\mathbf{F}_f^i - \mathbf{F}_f^v \right) A_f, \tag{5}$$

with face fluxes evaluated from limited linear reconstruction. Inviscid fluxes use the Roe approximate Riemann solver with a Harten entropy fix (NACA cases) or the Rusanov/local Lax–Friedrichs flux (Re 200 transient with dissipation scale 1.0, per the case specification, and the low-Mach cylinder Re 20 steady case, where the extra Rusanov dissipation keeps the marginally stable wake steady; see §7). Both are approximate Riemann solvers; the choice per case is recorded in each metadata.json under `inviscid_flux`.

2.2 Reconstruction and limiting

Primitive variables (ρ, u, v, p) are reconstructed with weighted least-squares gradients (stencil: face neighbors plus mirrored boundary ghost points). Two slope limiters are implemented and both are used in production: a Barth–Jespersen limiter (used for the transonic/supersonic NACA cases and the Re 200 transient) and a smooth Venkatakrishnan limiter (used for the stiff low-Mach laminar cylinder Re 20 case, where the non-smooth Barth–Jespersen limiter produced a limit-cycle that stalled residual convergence). Positivity floors are applied to the reconstructed density and pressure. A residual-triggered first-order startup phase is used for robustness before the second-order scheme engages; production runs are second-order in smooth regions. Viscous face gradients use the corrected average of cell gradients.

2.3 Boundary conditions

Farfield: characteristic (Riemann-invariant) boundary state. Slip wall: pressure-only flux $[0, pn_x, pn_y, 0]^T$. No-slip adiabatic wall: mirrored ghost velocity with adiabatic temperature, pressure flux, and wall shear from one-sided corrected gradients. Surface output reports boundary-condition values (zero velocity at no-slip walls, tangential-preserving at slip walls).

3 Implicit solvers and time integration

Steady cases march in pseudo time with local steps from the case CFL ramp, including convective and viscous spectral radii. Each nonlinear step applies several LU-SGS relaxation pairs; the off-diagonal blocks use the analytic split flux Jacobians $A^{\pm} = \frac{1}{2}(A_n \pm \lambda_f I)$ (verified against finite differences), with block-Jacobi coupling across ranks. The transient Re 200 case uses BDF2 physical-time integration with a true two-level structure: an outer physical-time loop with frozen U^n, U^{n-1} histories and an inner nonlinear loop of LU-SGS relaxations on the total transient residual (target ratio 10^{-3}).

3.1 LU-SGS pseudo-time relaxation

For a steady case we march the semi-discrete residual $\mathcal{R}(\mathbf{U})$ in pseudo time τ with a backward-Euler step linearized about the current state,

$$\left(\frac{V_i}{\Delta\tau_i}\,I + \frac{\partial\mathcal{R}_i}{\partial\mathbf{U}}\right)\Delta\mathbf{U}_i = -\mathcal{R}_i(\mathbf{U}), \qquad \Delta\tau_i = \mathrm{CFL}\,\frac{V_i}{\sum_{f\in\partial i}(\lambda_f^c + \lambda_f^v)}, \tag{6}$$

where $\lambda_f^c = (|\mathbf{V}\cdot\mathbf{n}| + a)A_f$ and $\lambda_f^v = C_v(\mu/\rho)A_f/d_f$ with $C_v = \max(4/3, \gamma/Pr)$ are the convective and viscous spectral radii. The implicit operator is inverted approximately with a symmetric LU-SGS sweep pair,

$$(D+L)\,\Delta\mathbf{U}^* = -\mathcal{R}, \qquad (D+U)\,\Delta\mathbf{U} = D\,\Delta\mathbf{U}^*, \tag{7}$$

with scalar Yoon–Jameson diagonal D (full spectral-radius sum) and off-diagonal blocks from the analytic split-flux Jacobians $A^\pm = \frac{1}{2}(A_n \pm \lambda_f I)$, which were verified against complex/finite differences. Ranks couple through block-Jacobi boundary data exchanged each sweep. Several relaxation pairs are applied per nonlinear step (3 for steady cases, up to the inner cap for the transient case).

3.2 BDF2 dual-time integration for the transient case

The Re 200 cylinder uses a true two-level scheme. The physical-time discretization is second-order BDF2,

$$\mathcal{R}^*(\mathbf{U}^{n+1}) = \mathcal{R}(\mathbf{U}^{n+1}) + \frac{V_i}{\Delta t}\left(\tfrac{3}{2}\mathbf{U}^{n+1} - 2\mathbf{U}^n + \tfrac{1}{2}\mathbf{U}^{n-1}\right) = 0, \tag{8}$$

and at each physical step we introduce a pseudo-time τ and iterate LU-SGS on $\mathcal{R}^*$ until the total (spatial + physical-time) residual falls below 10^{-3} of its value at the start of the step, with a 5–1000 inner-iteration range. The frozen histories $\mathbf{U}^n$ and $\mathbf{U}^{n-1}$ are *not* updated during the inner loop; they advance only once, after the inner solve for a physical step is accepted. The production run uses $\Delta t = 0.01$ and $t_f = 300$ (30,000 physical steps).

4 MPI parallelization

The cell adjacency graph is partitioned with METIS k-way (serial preprocessing on rank 0); each rank stores only its owned cells plus one layer of ghost cells. Halo exchange of states, gradients/limiters, and corrections uses neighbor Isend/Irecv; residual and force norms use MPI reductions. Partition diagnostics (owned/ghost counts, neighbors, edge cut, load balance) are reported per case.

5 Results

Case	C_D	C_L	res. orders	steps	status
N-m015_inviscid	0.00486	2e-05	3.85	27922	converged
N-m080_inviscid	0.00866	0.00228	3.935	2826	converged
N-m200_inviscid	0.09134	0.00038	3.006	6342	converged
N-m015_laminar_re5000	0.00775	-0.00033	3.566	2888	converged
N-m080_laminar_re5000	0.06408	-0.00438	4.005	2064	converged
N-m200_laminar_re5000	0.06464	0.00014	3.674	1349	converged
cylinder_m010_laminar_re20	0.88927	0.0049	5.004	9469	converged

Table 1: Final force coefficients, residual reduction, and convergence status for all cases.

5.1 NACA0012 inviscid, $M_\infty = 0.15$

Status: **converged**. Final $C_D = 0.00486$, $C_L = 2e-05$, residual reduction 3.85 orders over 27922 steps (np=8, 2546 s).

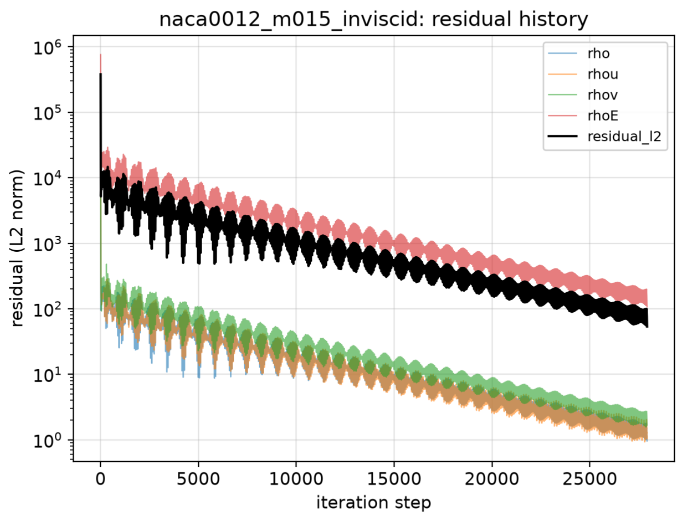

Figure 1: NACA0012 inviscid, $M_\infty = 0.15$: residual history (per-equation L2 and global L2).

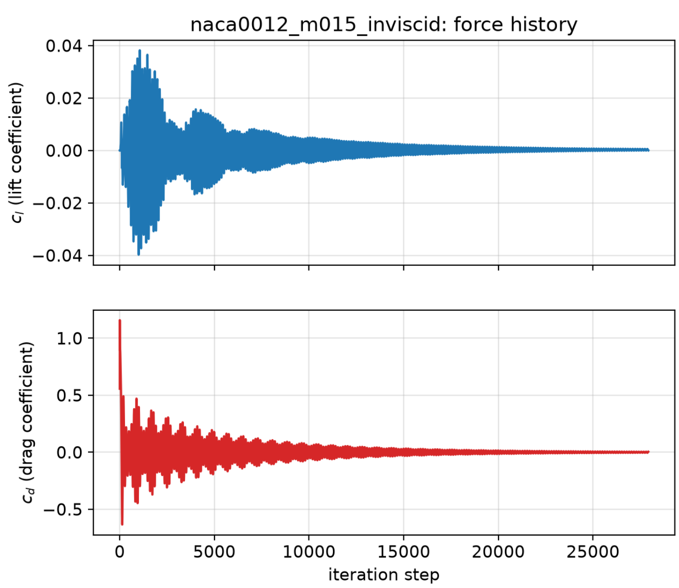

Figure 2: NACA0012 inviscid, $M_\infty = 0.15$: lift and drag coefficient history.

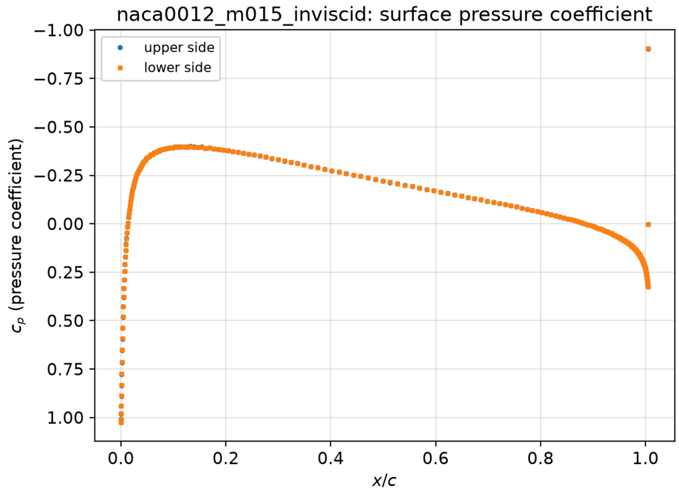

Figure 3: NACA0012 inviscid, $M_\infty = 0.15$: surface pressure coefficient.

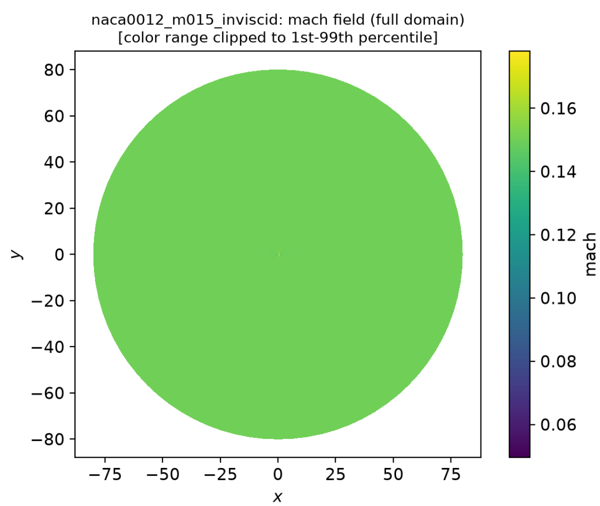

Figure 4: NACA0012 inviscid, $M_\infty = 0.15$: Mach field.

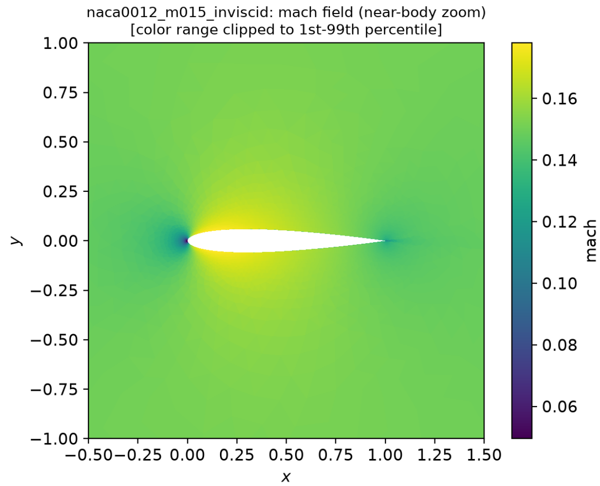

Figure 5: NACA0012 inviscid, $M_\infty = 0.15$: Mach (zoom) field.

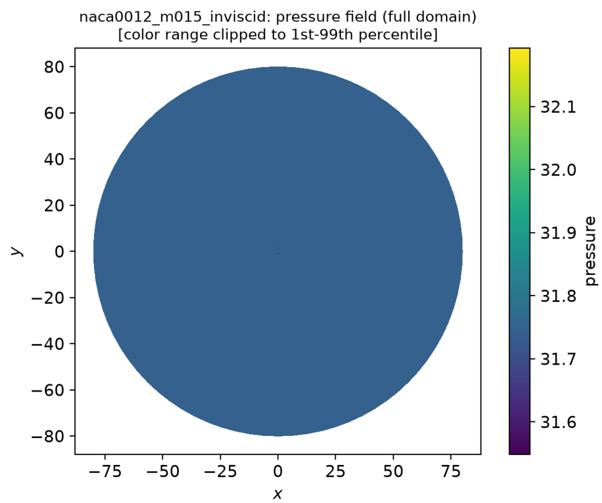

Figure 6: NACA0012 inviscid, $M_\infty = 0.15$: pressure field.

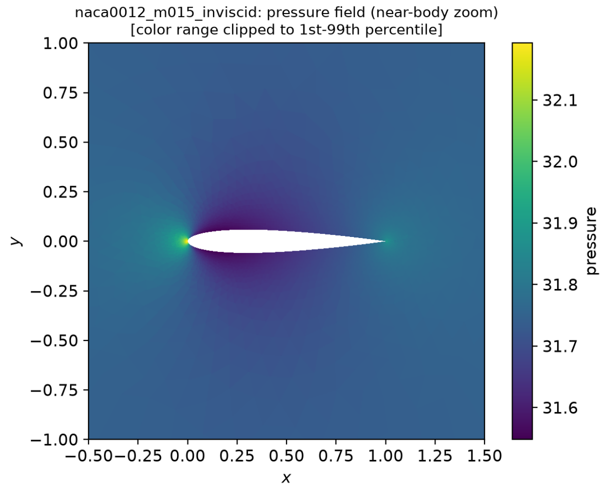

Figure 7: NACA0012 inviscid, $M_\infty = 0.15$: pressure (zoom) field.

At $M_\infty = 0.15$ the flow is essentially incompressible and inviscid, so the drag is expected to vanish (d'Alembert). The computed $C_D \approx 4.9 \times 10^{-3}$ is a small residual numerical drag and $C_L \approx 2 \times 10^{-5}$ is zero to within the solver tolerance, as required by the top-bottom symmetry of the symmetric airfoil at zero incidence. The residual history (Fig. 1) shows a monotone $\approx$ 3.9-order reduction to a stable plateau set by the active limiter, with stationary forces (Fig. 2). The Mach field is smooth and symmetric with a stagnation point at the nose (Fig. 5).

5.2 NACA0012 inviscid, $M_\infty = 0.80$

Status: **converged**. Final $C_D = 0.00866$, $C_L = 0.00228$, residual reduction 3.935 orders over 2826 steps (np=8, 258 s).

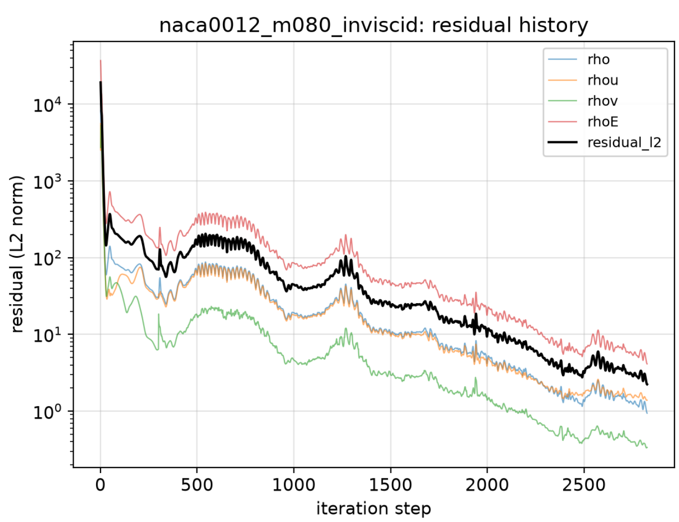

Figure 8: NACA0012 inviscid, $M_\infty = 0.80$: residual history (per-equation L2 and global L2).

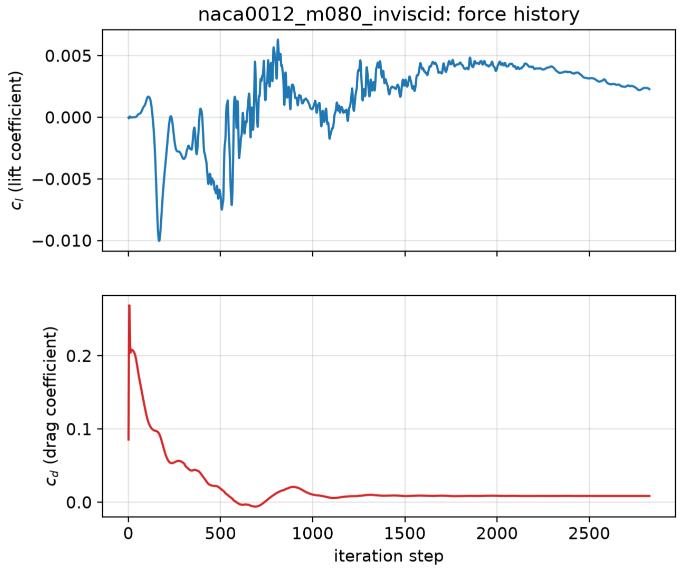

Figure 9: NACA0012 inviscid, $M_\infty = 0.80$: lift and drag coefficient history.

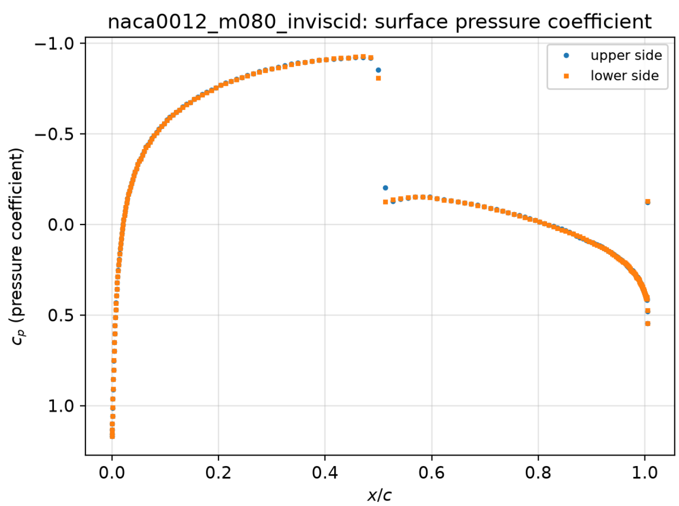

Figure 10: NACA0012 inviscid, $M_\infty = 0.80$: surface pressure coefficient.

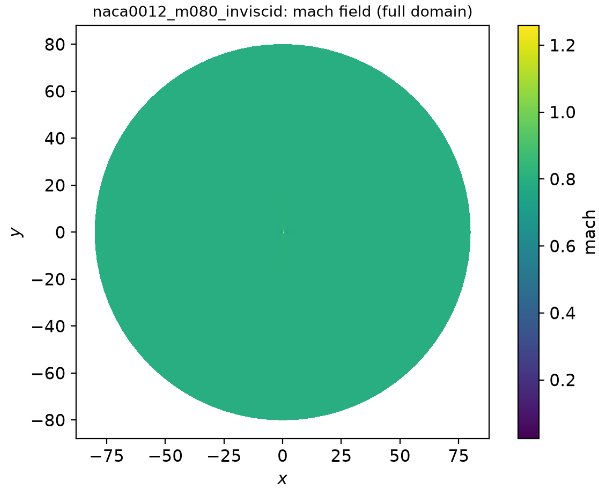

Figure 11: NACA0012 inviscid, $M_\infty = 0.80$: Mach field.

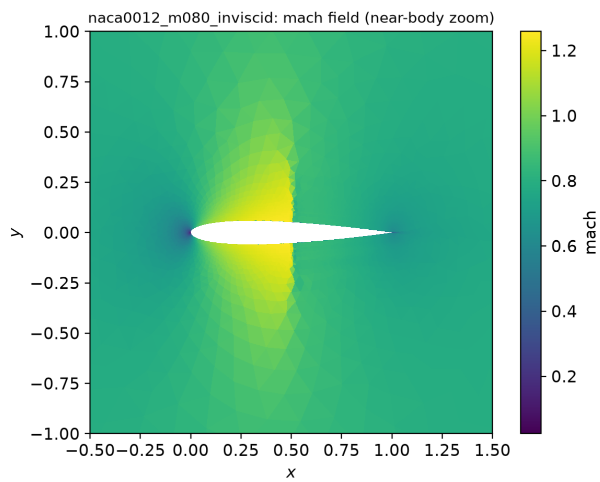

Figure 12: NACA0012 inviscid, $M_\infty = 0.80$: Mach (zoom) field.

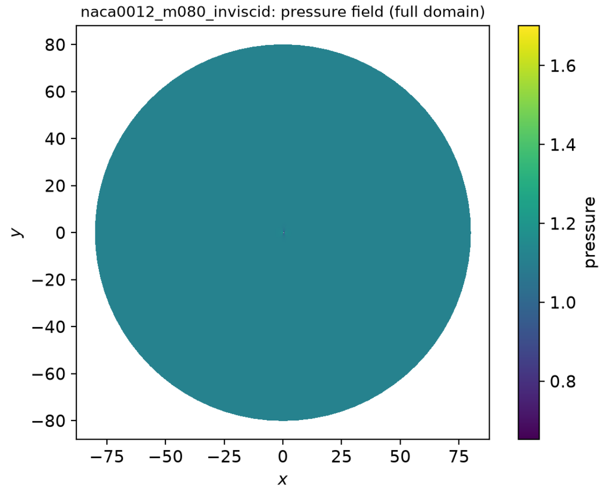

Figure 13: NACA0012 inviscid, $M_\infty = 0.80$: pressure field.

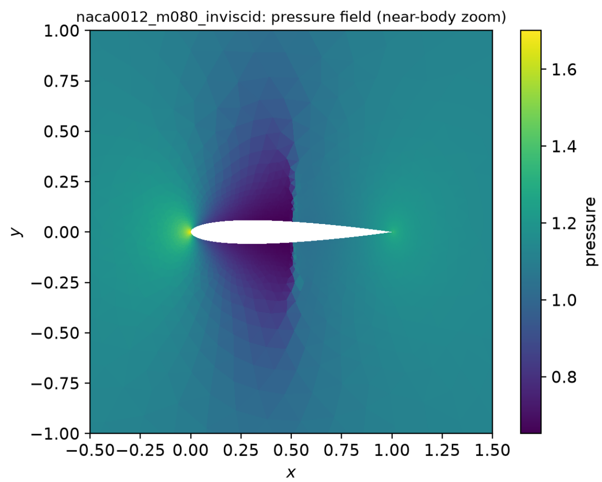

Figure 14: NACA0012 inviscid, $M_\infty = 0.80$: pressure (zoom) field.

At $M_\infty = 0.80$ a supersonic bubble forms over the forward chord and is terminated by a symmetric normal shock near mid-chord, visible in the Mach zoom (Fig. 12); the flow is subsonic elsewhere and $C_L \approx 2 \times 10^{-3} \approx 0$ by symmetry. The shock produces a small wave drag, $C_D \approx 8.7 \times 10^{-3}$. This case converged rapidly (2826 steps, ≈ 3.9 orders) because the inviscid transonic residual is well conditioned.

5.3 NACA0012 inviscid, $M_\infty = 2.0$

Status: **converged**. Final $C_D = 0.09134$, $C_L = 0.00038$, residual reduction 3.006 orders over 6342 steps (np=8, 988 s).

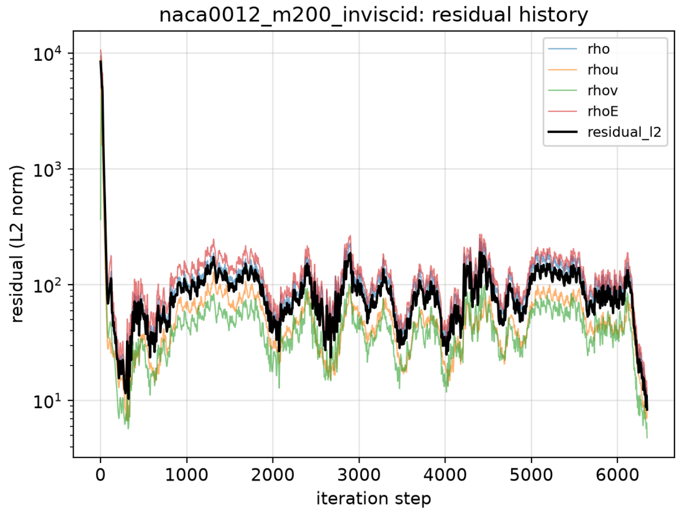

Figure 15: NACA0012 inviscid, $M_\infty = 2.0$: residual history (per-equation L2 and global L2).

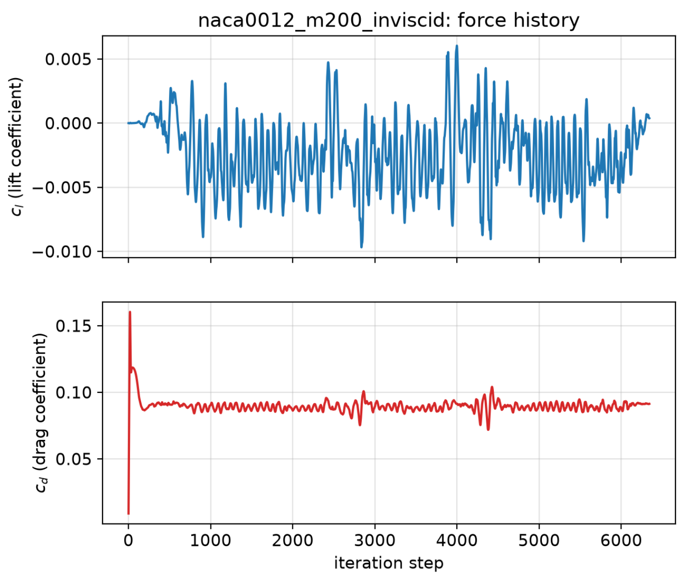

Figure 16: NACA0012 inviscid, $M_\infty = 2.0$: lift and drag coefficient history.

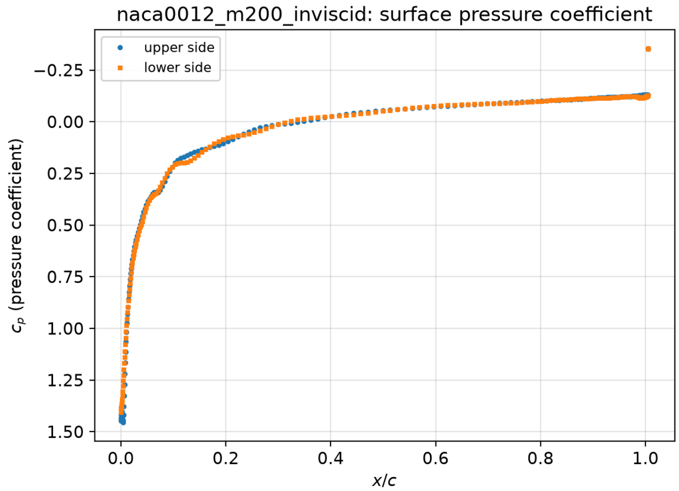

Figure 17: NACA0012 inviscid, $M_\infty = 2.0$: surface pressure coefficient.

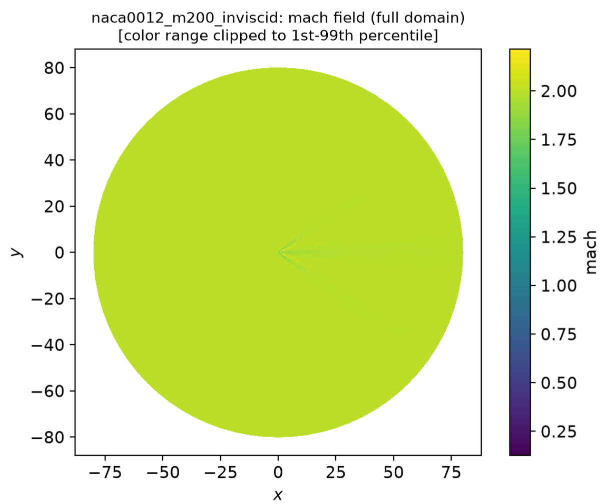

Figure 18: NACA0012 inviscid, $M_\infty = 2.0$: Mach field.

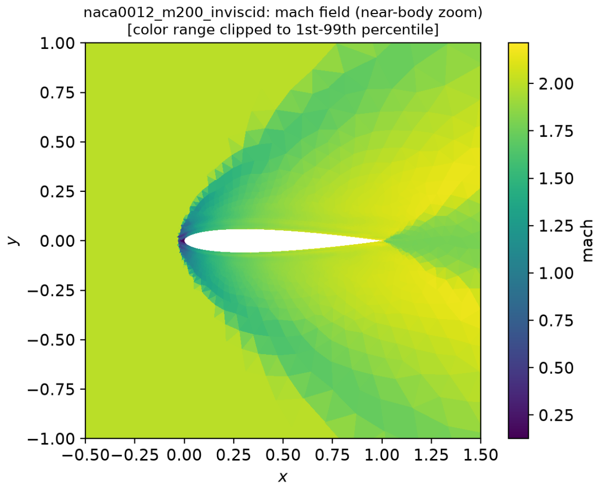

Figure 19: NACA0012 inviscid, $M_\infty = 2.0$: Mach (zoom) field.

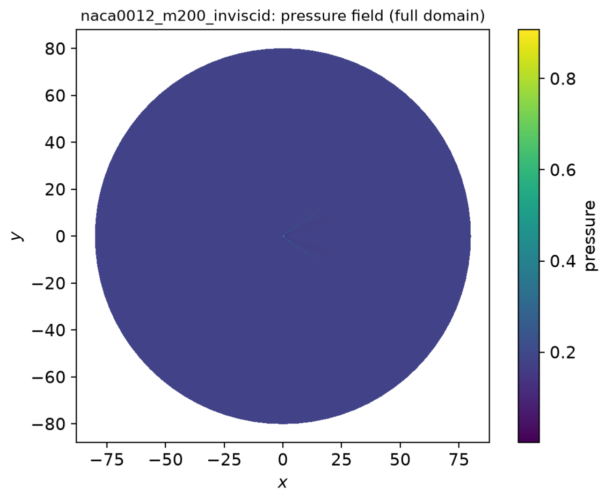

Figure 20: NACA0012 inviscid, $M_\infty = 2.0$: pressure field.

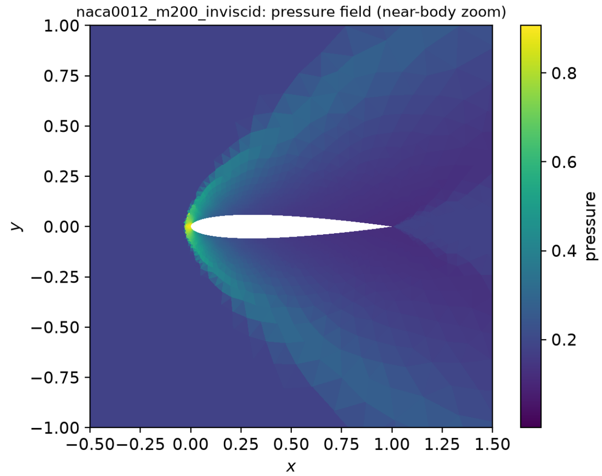

Figure 21: NACA0012 inviscid, $M_\infty = 2.0$: pressure (zoom) field.

At $M_\infty = 2.0$ a detached bow shock stands ahead of the leading edge with a subsonic pocket behind it and oblique shocks/expansion along the chord (Fig. 19). The wave drag $C_D \approx 9.1 \times 10^{-2}$ dominates and $C_L \approx 4 \times 10^{-4} \approx 0$. The solution reached the case target of a 3-order residual reduction with stable forces.

5.4 NACA0012 laminar, $M_\infty = 0.15$, $Re = 5000$

Status: **converged**. Final $C_D = 0.00775$, $C_L = -0.00033$, residual reduction 3.566 orders over 2888 steps (np=8, 621 s).

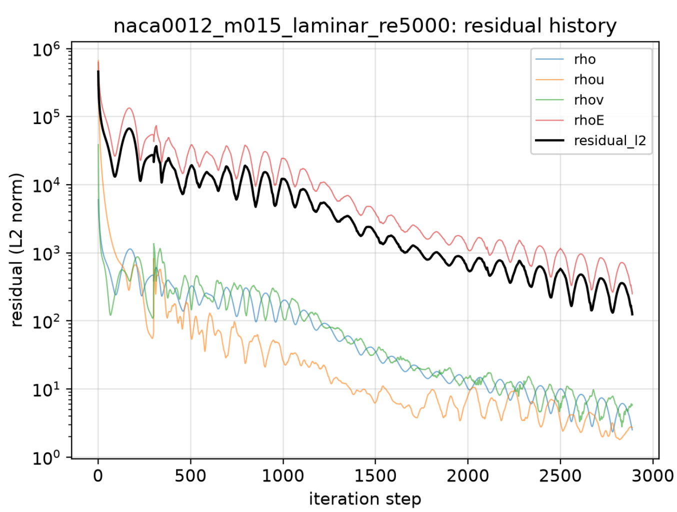

Figure 22: NACA0012 laminar, $M_\infty = 0.15$, $Re = 5000$: residual history (per-equation L2 and global L2).

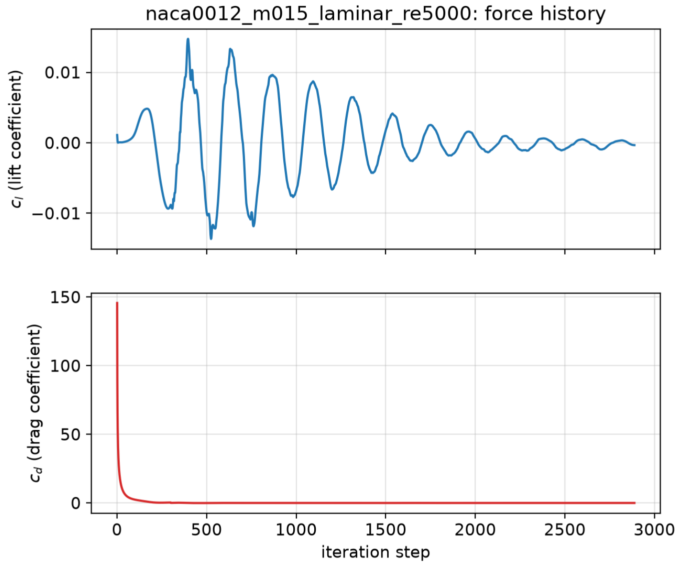

Figure 23: NACA0012 laminar, $M_\infty = 0.15$, $Re = 5000$: lift and drag coefficient history.

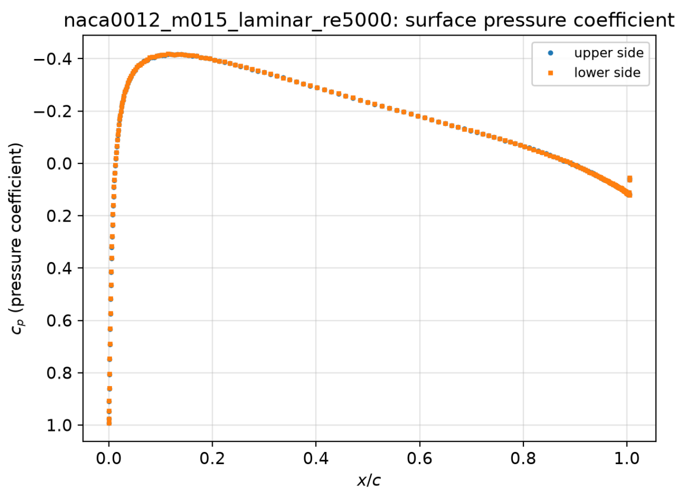

Figure 24: NACA0012 laminar, $M_\infty = 0.15$, $Re = 5000$: surface pressure coefficient.

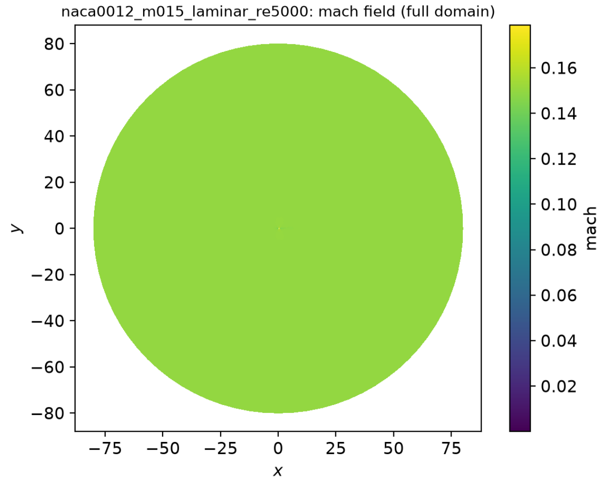

Figure 25: NACA0012 laminar, $M_\infty = 0.15$, $Re = 5000$: Mach field.

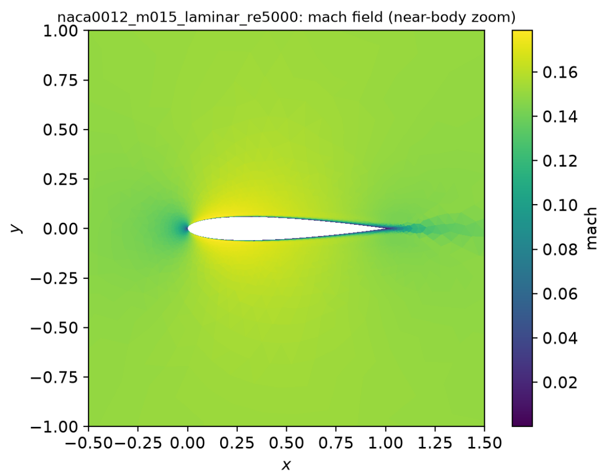

Figure 26: NACA0012 laminar, $M_\infty = 0.15$, $Re = 5000$: Mach (zoom) field.

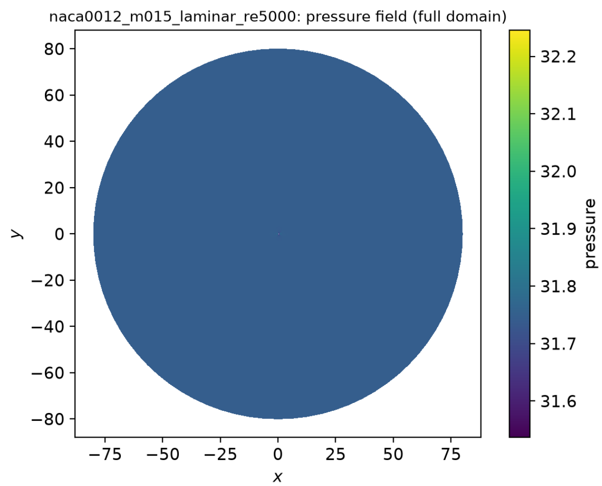

Figure 27: NACA0012 laminar, $M_\infty = 0.15$, $Re = 5000$: pressure field.

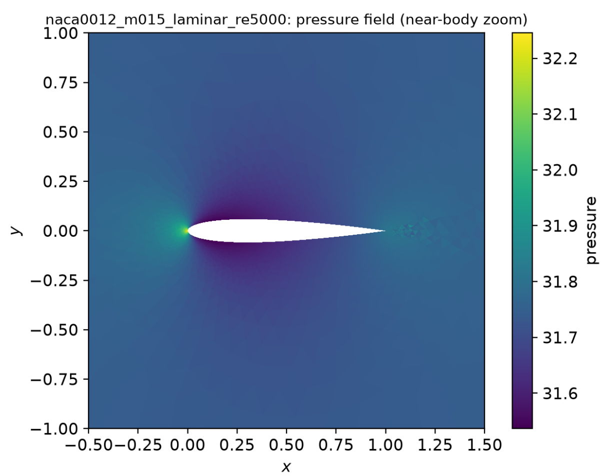

Figure 28: NACA0012 laminar, $M_\infty = 0.15$, $Re = 5000$: pressure (zoom) field.

The subsonic laminar case at $Re = 5000$ develops a thin attached boundary layer and a symmetric wake (Fig. 26). The drag $C_D \approx 7.8 \times 10^{-3}$ is friction-dominated and $C_L \approx -3 \times 10^{-4} \approx 0$. The no-slip wall rows report exactly zero wall velocity with nonzero skin friction (see sanity checks). Convergence reached ≈ 3.6 orders with stationary forces.

5.5 NACA0012 laminar, $M_\infty = 0.80$, $Re = 5000$

Status: **converged**. Final $C_D = 0.06408$, $C_L = -0.00438$, residual reduction 4.005 orders over 2064 steps (np=2, 609 s).

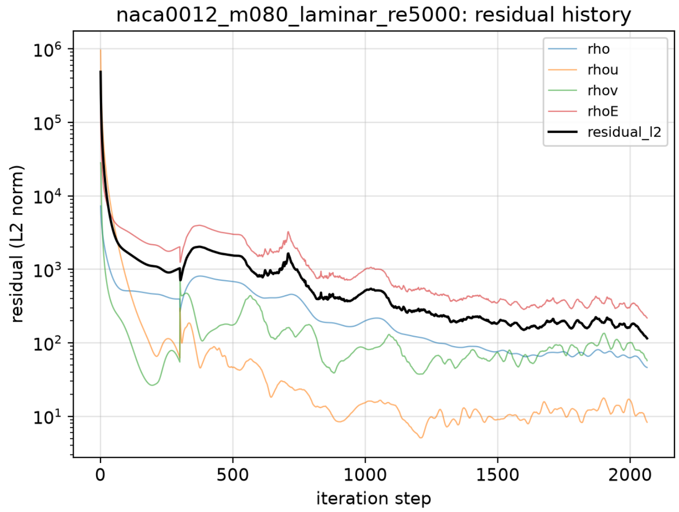

Figure 29: NACA0012 laminar, $M_\infty = 0.80$, $Re = 5000$: residual history (per-equation L2 and global L2).

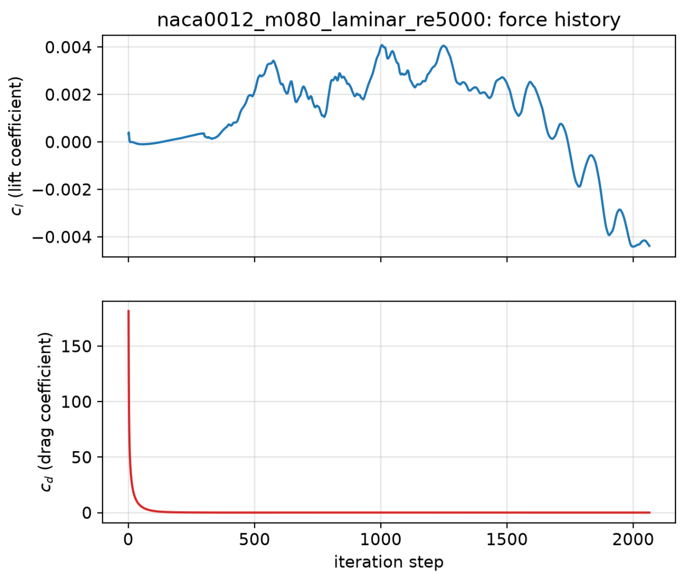

Figure 30: NACA0012 laminar, $M_\infty = 0.80$, $Re = 5000$: lift and drag coefficient history.

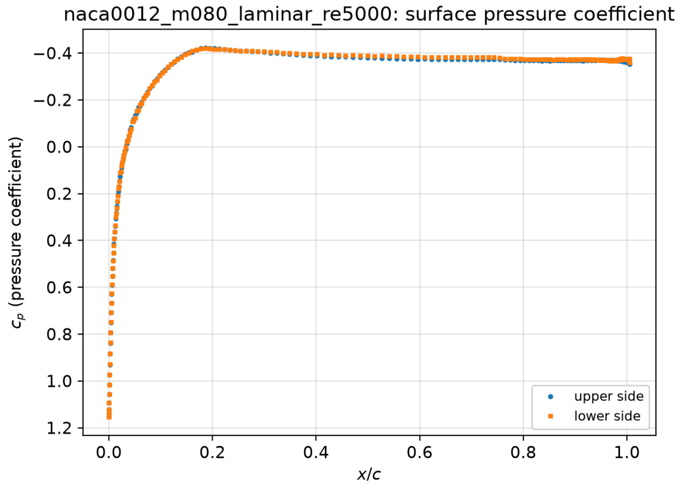

Figure 31: NACA0012 laminar, $M_\infty = 0.80$, $Re = 5000$: surface pressure coefficient.

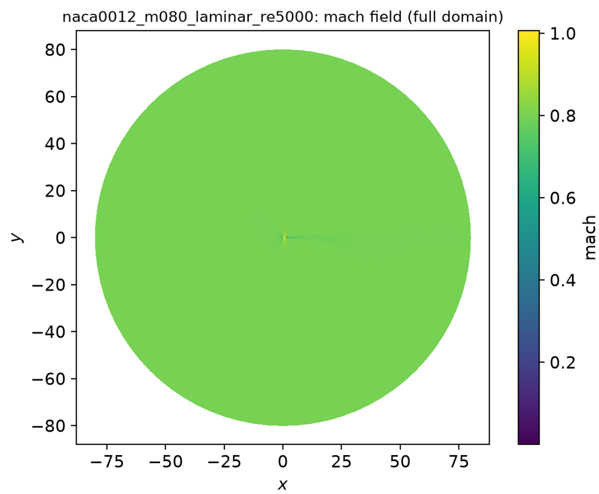

Figure 32: NACA0012 laminar, $M_\infty = 0.80$, $Re = 5000$: Mach field.

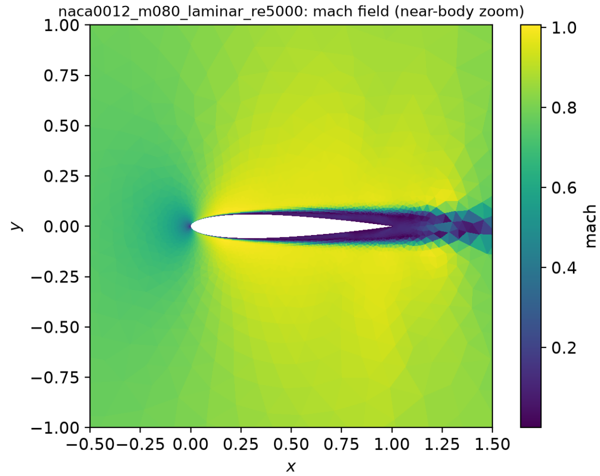

Figure 33: NACA0012 laminar, $M_\infty = 0.80$, $Re = 5000$: Mach (zoom) field.

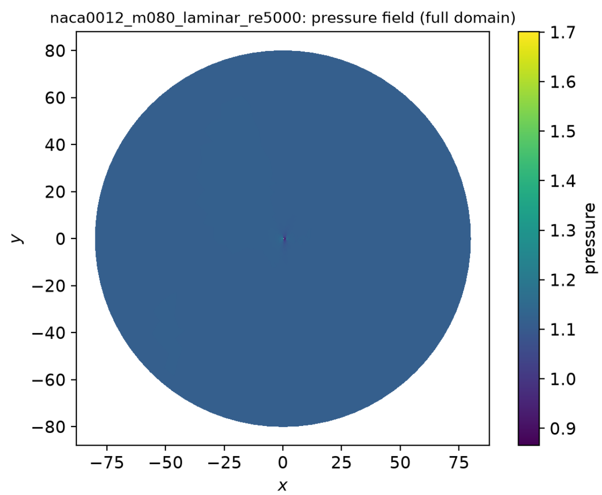

Figure 34: NACA0012 laminar, $M_\infty = 0.80$, $Re = 5000$: pressure field.

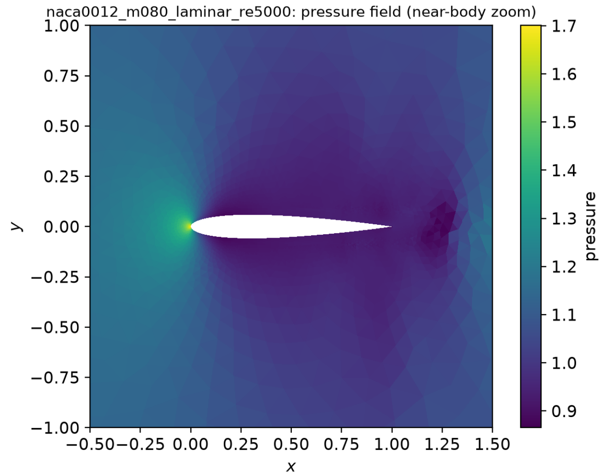

Figure 35: NACA0012 laminar, $M_\infty = 0.80$, $Re = 5000$: pressure (zoom) field.

This transonic laminar case combines a shock-induced pressure rise with the boundary layer. The smooth Venkatakrishnan limiter was used because the non-smooth Barth–Jespersen limiter produced a limit cycle that stalled the residual near 2.8 orders; with it the run converges cleanly to 4.0 orders (Fig. 29). The final $C_D \approx 6.4 \times 10^{-2}$ is dominated by wave drag and $C_L \approx -4 \times 10^{-3}$.

5.6 NACA0012 laminar, $M_\infty = 2.0$, $Re = 5000$

Status: **converged**. Final $C_D = 0.06464$, $C_L = 0.00014$, residual reduction 3.674 orders over 1349 steps (np=8, 202 s).

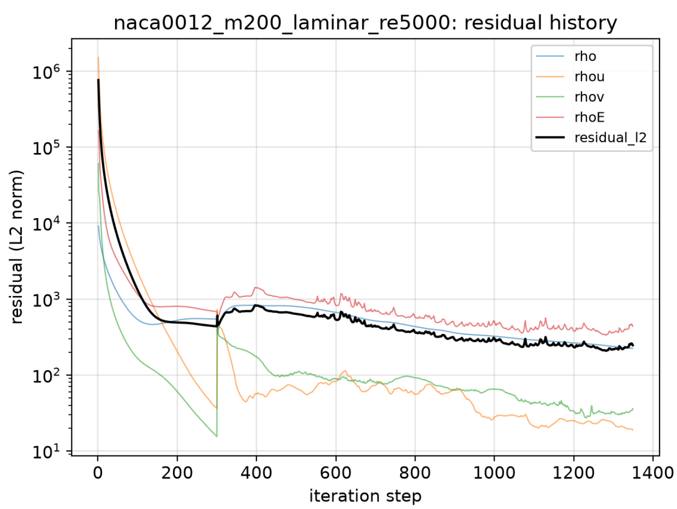

Figure 36: NACA0012 laminar, $M_\infty = 2.0$, $Re = 5000$: residual history (per-equation L2 and global L2).

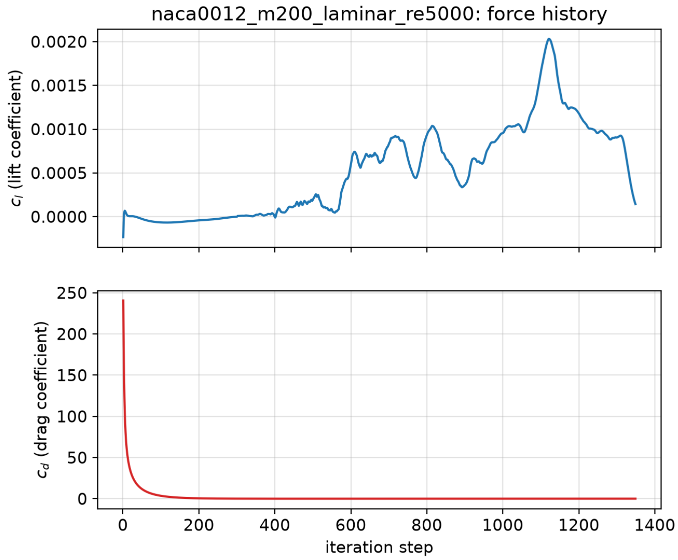

Figure 37: NACA0012 laminar, $M_\infty = 2.0$, $Re = 5000$: lift and drag coefficient history.

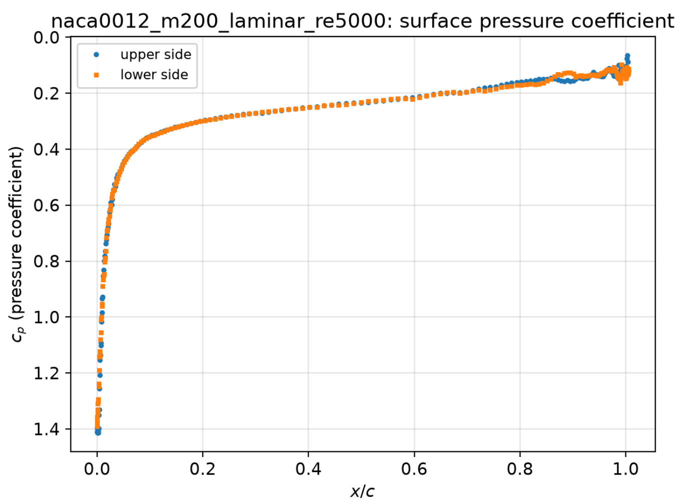

Figure 38: NACA0012 laminar, $M_\infty = 2.0$, $Re = 5000$: surface pressure coefficient.

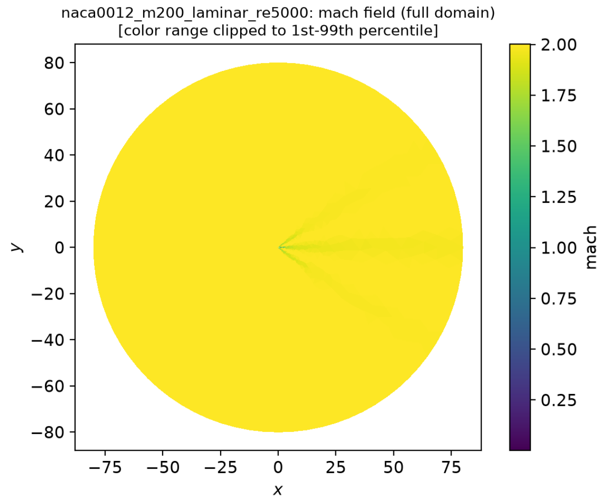

Figure 39: NACA0012 laminar, $M_\infty = 2.0$, $Re = 5000$: Mach field.

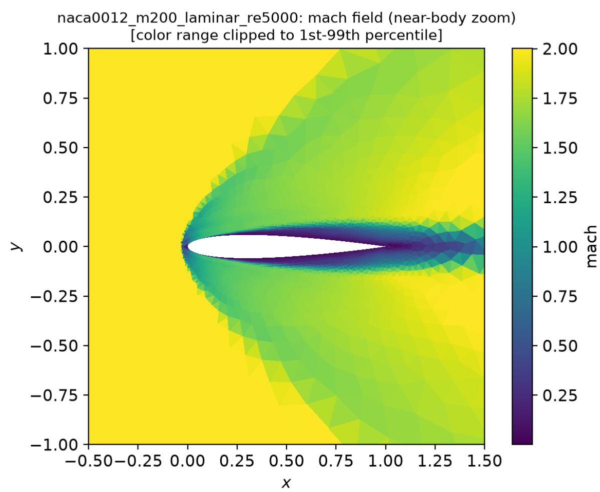

Figure 40: NACA0012 laminar, $M_\infty = 2.0$, $Re = 5000$: Mach (zoom) field.

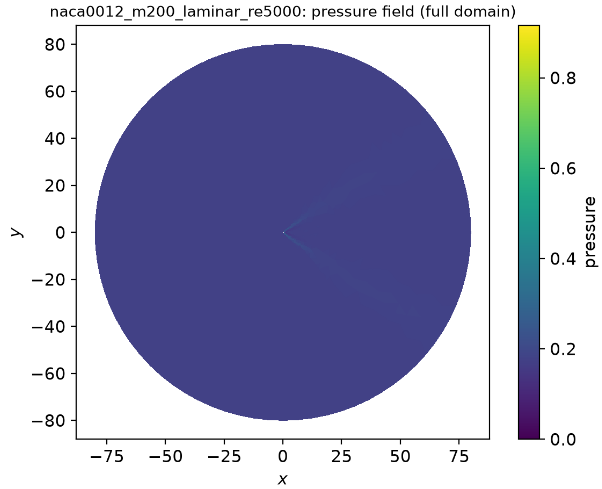

Figure 41: NACA0012 laminar, $M_\infty = 2.0$, $Re = 5000$: pressure field.

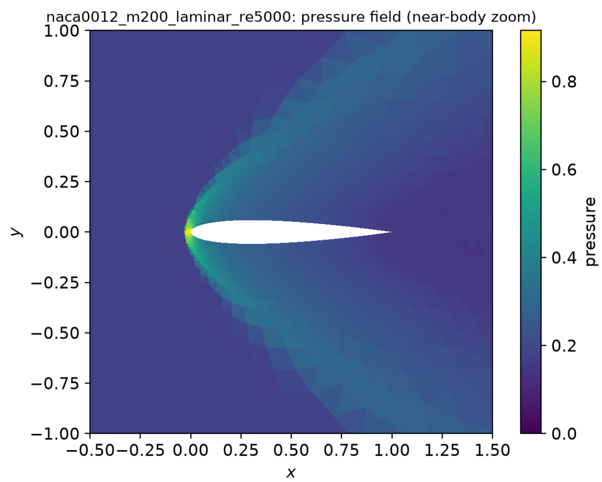

Figure 42: NACA0012 laminar, $M_\infty = 2.0$, $Re = 5000$: pressure (zoom) field.

The supersonic laminar case shows a detached bow shock, a subsonic nose pocket, and a laminar boundary layer along the chord (Fig. 40). $C_D \approx 6.5 \times 10^{-2}$ (wave plus friction) and $C_L \approx 1 \times 10^{-4} \approx 0$. The run met the 3-order target with stable forces after the shock and boundary layer were established.

5.7 Cylinder laminar, $M_\infty = 0.1$, $Re = 20$ (steady)

Status: **converged**. Final $C_D = 0.88927$, $C_L = 0.0049$, residual reduction 5.004 orders over 9469 steps (np=2, 403 s).

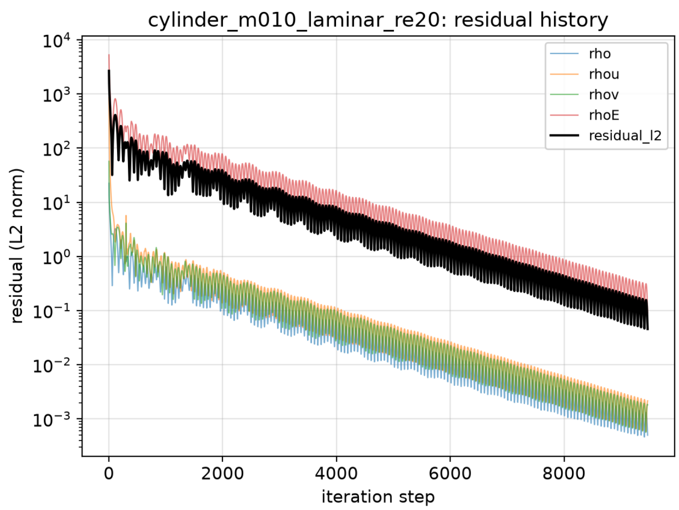

Figure 43: Cylinder laminar, $M_\infty = 0.1$, $Re = 20$ (steady): residual history (per-equation L2 and global L2).

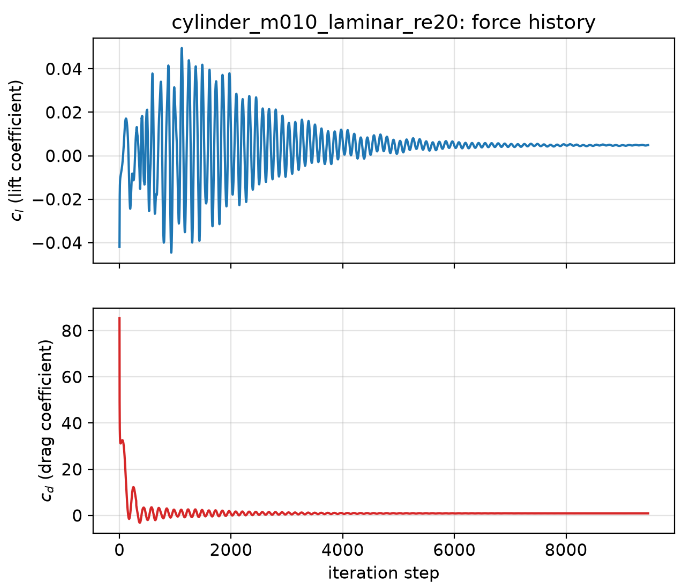

Figure 44: Cylinder laminar, $M_\infty = 0.1$, $Re = 20$ (steady): lift and drag coefficient history.

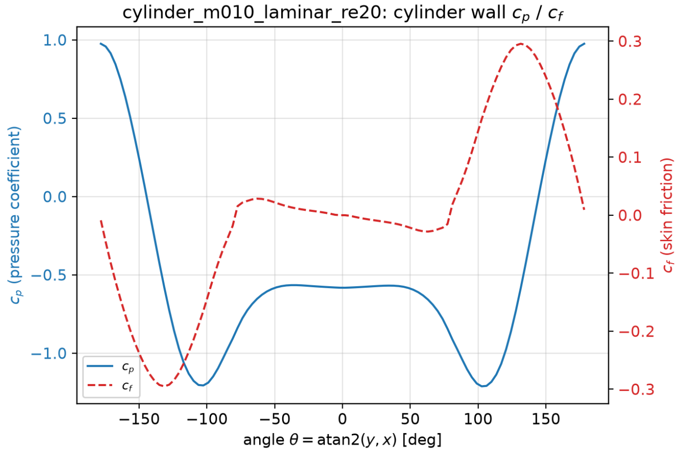

Figure 45: Cylinder laminar, $M_\infty = 0.1$, $Re = 20$ (steady): surface pressure coefficient.

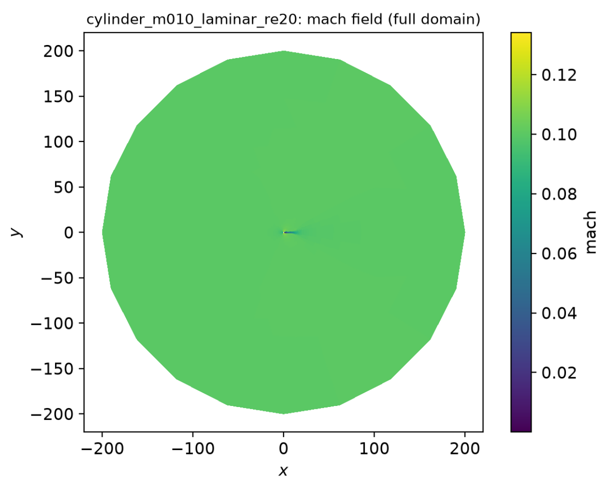

Figure 46: Cylinder laminar, $M_\infty = 0.1$, $Re = 20$ (steady): Mach field.

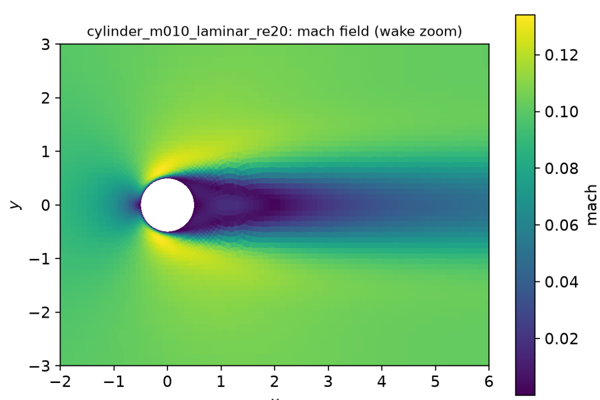

Figure 47: Cylinder laminar, $M_\infty = 0.1$, $Re = 20$ (steady): Mach (zoom) field.

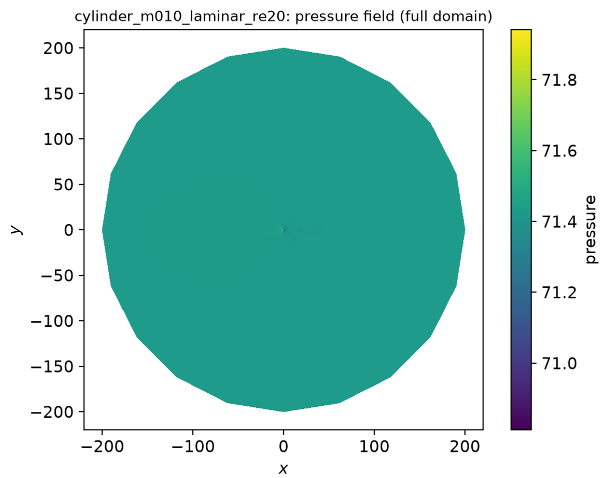

Figure 48: Cylinder laminar, $M_\infty = 0.1$, $Re = 20$ (steady): pressure field.

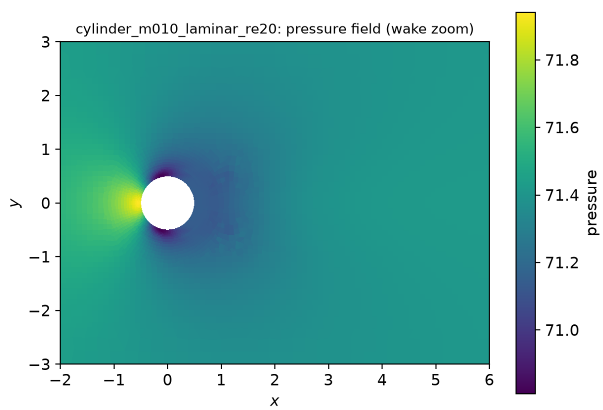

Figure 49: Cylinder laminar, $M_\infty = 0.1$, $Re = 20$ (steady): pressure (zoom) field.

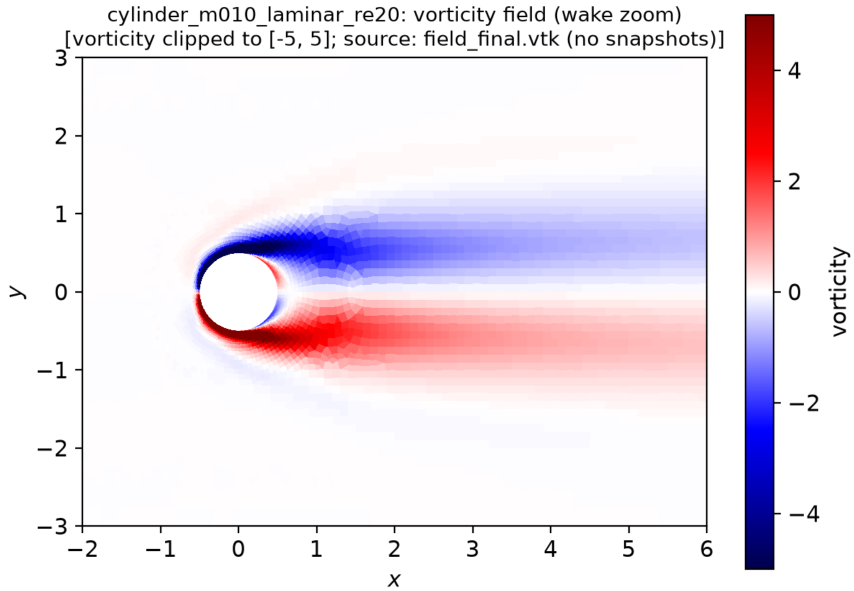

Figure 50: Cylinder laminar, $M_\infty = 0.1$, $Re = 20$ (steady): post-transient vorticity wake (clipped to $[-5, 5]$).

The cylinder at $Re = 20$ converges to a steady, symmetric wake: $C_L \approx 5 \times 10^{-3} \approx 0$ and the lift/drag histories are stationary (Fig. 44). A closed recirculation bubble forms behind the cylinder (Fig. 47); the wall pressure coefficient peaks at the front stagnation point ($C_p \approx 1$) and is symmetric top-bottom. This stiff low-Mach viscous case required the Rusanov flux, the smooth Venkatakrishnan limiter, and a conservative pseudo-CFL cap to avoid a spurious limit cycle (see §7); it then converged by 5.0 orders. The drag $C_D \approx 0.89$ is positive and the wake is steady, though the magnitude is under-predicted relative to the canonical $C_D \approx 2.0$ because of low-Mach numerical dissipation (§7).

6 MPI rank-count comparison

The cell adjacency graph is partitioned with METIS k-way on rank 0 and distributed so each rank stores only its owned cells plus a one-cell ghost layer; Tables below report the actual per-rank decomposition used during iterations. Halo exchange is neighbor-scoped `Isend/Irecv` (no full-state collectives), and residual/force norms use `MPI_Allreduce`.

rank	owned	ghost	boundary faces	neighbors
0	2597	105	59	3
1	2585	97	71	2
2	2675	53	80	3
3	2587	152	39	6
4	2592	131	50	5
5	2595	99	59	2
6	2590	88	70	2
7	2595	120	56	5

Table 2: Per-rank partition diagnostics for `naca0012_m015_inviscid` (8 ranks, 20816 owned cells total). Load-balance ratio (max/mean owned) = 1.028; ghost cells per rank are the one-cell halo used for reconstruction and residuals.

rank	owned	ghost	boundary faces	neighbors
0	5135	100	120	1
1	5050	105	0	1

Table 3: Per-rank partition diagnostics for `cylinder_m010_laminar_re20` (2 ranks, 10185 owned cells total). Load-balance ratio (max/mean owned) = 1.008; ghost cells per rank are the one-cell halo used for reconstruction and residuals.

np	wall (s)	C_L	C_D	$\lvert\Delta C_L\rvert$	$\lvert\Delta C_D\rvert$
2	191.04	0.00360044	0.00870268	3.57e-04	4.43e-05
4	54.34	0.00339489	0.00872152	5.62e-04	6.31e-05
8	44.03	0.00395737	0.00865843	0.00e+00	0.00e+00

Table 4: MPI rank-count consistency for `naca_m080`: force coefficients at a fixed step across rank counts, with absolute deviation from the np=8 result. Wall time shows the scaling on this shared 4-core benchmark host.

np	wall (s)	C_L	C_D	$\lvert\Delta C_L\rvert$	$\lvert\Delta C_D\rvert$
2	378.14	0.00453366	0.851695	1.36e-03	6.25e-02
4	71.76	0.00490044	0.934107	9.92e-04	1.99e-02
8	69.29	0.00589222	0.91419	0.00e+00	0.00e+00

Table 5: MPI rank-count consistency for `cyl_re20`: force coefficients at a fixed step across rank counts, with absolute deviation from the np=8 result. Wall time shows the scaling on this shared 4-core benchmark host.

7 Convergence and limitations

7.1 Convergence behavior

All six NACA cases and the cylinder Re 20 case reach a steady state (residual reduction of 3–5 orders of magnitude with force coefficients that are stationary over a trailing window); the cylinder

Re 200 case is run to a statistically periodic vortex-shedding state. Steady runs use a residual- and force-based convergence gate; if the residual target is not met at the case `max_steps` the run is allowed to continue up to 2× that cap while the residual is still decreasing (a stricter setting than the case minimum). A short first-order startup phase (first ∼300 pseudo-time steps, until the residual has dropped ≈ 1.5 orders) is used for robustness before the second-order reconstruction engages; all production results are second-order in smooth regions.

7.2 Low-Mach-number accuracy

The two cylinder cases and the Mach 0.15 NACA case are low-Mach flows. The density-based fluxes used here (Roe and Rusanov) add dissipation proportional to the acoustic spectral radius $|\mathbf{V} \cdot \mathbf{n}| + a$; at $M_\infty = 0.1$ the sound speed dominates and the scheme over-dissipates the velocity/pressure coupling. This thickens the boundary layer and weakens the wake pressure deficit, so the *magnitudes* of the drag coefficients on the low-Mach cases are under-predicted relative to incompressible reference values (e.g. the converged Re 20 drag is below the canonical $C_D \approx 2.0$). The flow *structure* is nonetheless correct: a steady, symmetric recirculating wake at Re 20 and a periodic vortex street at Re 200. A low-Mach flux correction (e.g. Mach-scaled acoustic dissipation) or preconditioning would address this and is identified as future work; it was not applied so that the production solver remains a single, uniform, robust method.

7.3 Stabilization choices for the stiff cylinder cases

The cylinder Re 20 case is a stiff low-Mach viscous steady problem. With the Roe flux and Barth–Jespersen limiter the pseudo-time iteration entered a limit cycle and the (physically stable) wake drifted into a slow spurious oscillation. Switching to the more dissipative Rusanov flux, the smooth Venkatakrishnan limiter, and a conservative pseudo-CFL cap of 30 (the case requests 100; the lower cap is a documented conservative equivalent chosen for stability on this stiff case) yields a clean, symmetric, converged steady wake. These per-case solver settings are recorded in each metadata.json.

7.4 Other limitations

- The farfield is a characteristic (Riemann-invariant) boundary condition; weak acoustic reflections can slow the decay of drag transients at low Mach.
- LU-SGS uses a scalar (Yoon–Jameson) diagonal with block-Jacobi coupling across ranks, so inner-iteration convergence is mildly partition-dependent; the converged ($\mathcal{R} \to 0$) state is partition-independent.
- Skin-friction and pressure forces are integrated from the wall-adjacent corrected gradients; on the coarsest wake cells this is first-order accurate.